Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: [Insert date]

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick “No” or write “None” – do not leave it blank. Submit as a single PDF document named “SDC2026_TeamName_TechDeclaration.pdf” together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

1. TEAM AND PROJECT OVERVIEW

Team Name

ZH

Project / Solution Name

ChaseFlow for Orbyte (working name)

Name (Leader)

Wong Zong Han

Email & Phone (Leader)

johnwong2403@gmail.com / 016-4974388

University Affiliation(s)

Asia Pacific University of Technology & Innovation (APU)

Live Prototype URL

https://chaseflow-orbyte.replit.app

Code Repository URL (public / access granted to judges)

https://github.com/johnwong2403/chaseflow-orbyte

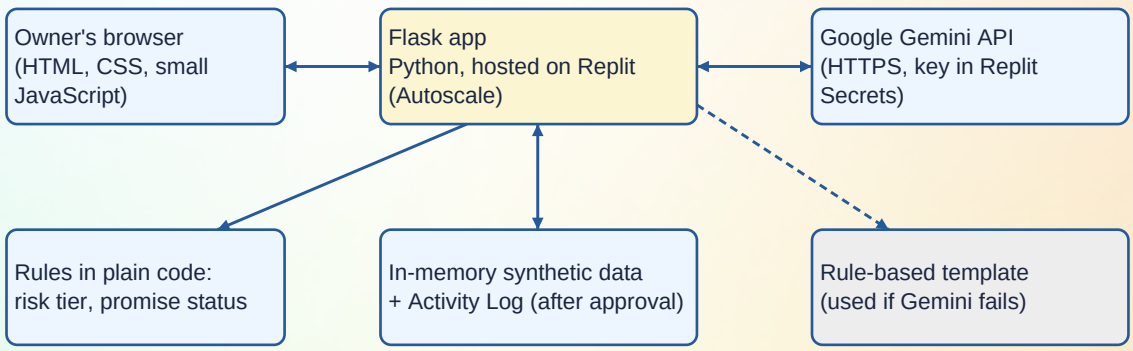
2. ARCHITECTURE & TECHNOLOGY STACK

2.1. System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

A browser shows server-rendered pages (HTML, CSS, small JavaScript) from a Python Flask backend hosted on Replit (Autoscale deployment). The backend keeps synthetic customers, invoices, an activity log and payment promises in memory. Plain-code rules turn this data into a risk tier and a status for each promise. When the owner opens an overdue invoice, the backend sends only the needed facts (customer name, invoice number, amount, days overdue, past late payments, promise status) to the Google Gemini API over HTTPS, using a key stored in Replit Secrets. The answer is validated, retried once if it fails, cached, and replaced by a rule-based template if the AI is unavailable. The owner edits and approves the draft; the approval is written to the Activity Log. Sending is simulated, so nothing leaves the app. There is no database, no GPU service and no other cloud service.



System block diagram. Sending is simulated: nothing leaves the app.

2. ARCHITECTURE & TECHNOLOGY STACK

2.2. Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.11.14	Application language	PSF License
Flask 3.1.3 with (Jinja2 3.1.6, Werkzeug 3.1.9, MarkupSafe 3.0.4, itsdangerous 2.2.0, click 8.5.0, blinker 1.9.0)	Web framework and HTML templates. The others are installed together with Flask	BSD-3-Clause (blinker: MIT)
requests 2.34.2	HTTPS calls to the Gemini API	Apache-2.0
Inter typeface (Google Fonts, loaded from the web)	Interface font	SIL Open Font License 1.1
Own code: HTML, CSS, vanilla JavaScript, inline SVG charts	Interface and charts. No front-end framework and no chart library	Apache-2.0 (this repository)
No other third-party libraries		None

3. APIs & CLOUD SERVICES

3.1. External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API and Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Gemini API (Generative Language API), Google	Explains the risk and drafts the reminder message in English, Bahasa Malaysia or Chinese	Free tier
Google Fonts, Google	Serves the Inter typeface to the browser	Free
No other external API	No API keys or secrets are included in this document	None

3. APIs & CLOUD SERVICES

3.2. Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit Workspace and Deployments (Autoscale), Replit	Editing and running the code, hosting the live prototype at chaseflow-orbyte.replit.app (limited to one machine), and keeping the API key in Replit Secrets	Replit Core (provided through the SDC portal). Region: Replit default
GitHub	Public source repository with commit history	Free
YouTube, Google	Hosts the 5 to 7 minute pitch video	Free
No other cloud service	No database service, no GPU service	None

4. AI TOOLS, MODELS & DATASETS

4.1. AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Gemini Flash-Lite (model name gemini-flash-lite-latest, Google's alias for its current version)	Google, through the Gemini API	Pre-trained, used as is. No fine-tuning and no self-trained model	Google Gemini API Terms of Service
No other AI model			None

4. AI TOOLS, MODELS & DATASETS

4.2. Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Demo customers and invoices	Written by the team: src/data.py in the repository	Synthetic (6 customers, 15 invoices)	Apache-2.0 (this repository)	N: names, phone numbers and amounts are made up
Demo reminders and payment promises	Written by the team: src/data.py in the repository	Synthetic (3 reminders, 2 promises)	Apache-2.0 (this repository)	N: made up
No other dataset		No training or fine-tuning data is used	None	

4. AI TOOLS, MODELS & DATASETS

4.3. AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
Claude (Anthropic) via claude.ai / Claude Code Sonnet 5.5	Wrote most of the Python, HTML, CSS and JavaScript code, the unit tests, the README, drafts of these documents and the logo drawing, following the team leader's instructions	Major
ChatGPT (OpenAI) GPT-6	A second opinion on product direction and priorities.	Minor
Replit Agent (Replit)	Ran once when the project was imported. It changed no files	Minor
Gemini (Google)	Inside the product only (see 4.1). Not used to write the project's code	Part of the product

4. AI TOOLS, MODELS & DATASETS

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

The team leader, an Accounting student interested in SME problems, chose the problem and scope; ideas were brainstormed with AI assistants. Each version was tested in the running app on Replit and the live site, and fixed until it worked. The core rules (risk tier, promise status, mark paid and undo, reset) are covered by 13 automated unit tests. AI output in the product is validated, shown next to the rule it is based on, and approved by a human before any action. The team leader is responsible for all submitted code and will explain it.

5. SPONSOR-PROVIDED RESOURCES

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Replit (SDC Hackathon sponsor-provided)	Replit Core subscription (provided through the SDC portal and used)	Development workspace and hosting of the live prototype
RunPod (SDC Hackathon sponsor-provided)	GPU credit: not used	Not used. The project has no workload that needs a GPU

5. SPONSOR-PROVIDED RESOURCES

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit is required by the competition and hosts the live prototype, but the app is a standard Flask program that could run on any Python host, so the platform itself could be replaced with little work. RunPod was not needed because the only AI component is the Gemini API, which runs on Google's servers and uses no GPU of ours.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.1. Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The app collects no real user data. All customers, invoices and promises are synthetic and were written by the team. The Gemini API receives only the customer's (made-up) business name, invoice number, amount, days overdue, late-payment count and promise status. Data lives in server memory and is erased when the server restarts or the demo is reset; no database or log of personal data is kept. The browser stores only the light/dark theme choice. The API key is kept in Replit Secrets, not in the code. In a real deployment the app would follow Malaysia's PDPA: collect only what collection needs, restrict access by role, limit retention, and send the AI minimal data.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.2. Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Hallucination or a wrong risk tier: a plain rule gives the baseline, the AI may move it by one level and must explain why, its JSON output is validated, and a human approves every message. Rate limits and outages: one automatic retry, drafts cached, then a fixed rule-based template. Dependence on one provider: the fallback keeps the app usable without Gemini. Translation quality may vary, so the owner can edit before approval. Demo data resets on restart because there is no database yet. The app never sends messages or processes payments, and gives no legal or financial advice.

7. COMPLIANCE & READINESS

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. DECLARATION

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature]

WONG

[Date]

23/10/26